Velocity	c	Special relativity	A maximum velocity exists
Information (actions)	$\hbar$	Quantum mechanics	A minimum of information exists
Length	Lp	Quantum gravity	A minimum length exists

There is another infinity which disorientates our thinking: the infinite spatial extension of the cosmos. But as I illustrated in Chapter 3, Einstein has found the way of thinking of a finite cosmos without borders. Current measurements indicate that the size of the cosmos must be larger than 100 billion light years. This is the order of magnitude of the universe we have indirect access to. It is around 10^{120} times greater than the Planck length, a number of times which is given by a 1 followed by 120 zeroes. Between the Planck scale and the cosmological one, then, there is the mind-blowing separation of 120 orders of magnitude. Huge. Extraordinarily huge. But *finite*.

In this space – between the size of the minute quanta of space, up to quarks, protons, atoms, chemical structures, mountains, stars, galaxies (each formed by one hundred billion stars), clusters of galaxies, and right up until the seemingly boundless visible universe of more than 100 billion galaxies – unfolds the swarming complexity of our universe; a universe we know only in a few aspects. Immense. Finite.

The cosmological scale is reflected in the value of the cosmological constant Λ, which enters into the basic equations of our theories. The fundamental theory contains, therefore, a very large number: the ratio between the cosmological constant and the Planck length. It is this large number that opens the way to the vast complexity of the world. But what we see and understand of the universe is not an infinity to drown in. It is a wide sea, but a finite one.

The book of *Ecclesiasticus*, or Sirach,[fn47] opens with a stupendous question:

> Who can number the sand of the sea, and the drops of rain, and the days of eternity? Who can find out the height of heaven, and the breadth of the earth, and the deep, and wisdom?

Not much longer after these lines were composed, another great text was written, with an opening which still resounds:

> Some think, O King Hiero, that the grains of sand cannot be counted.

This is the opening of *Psammites* (*The Sand Reckoner*) by Archimedes, in which the greatest scientist of antiquity … counts the grains of sand in the universe!